

The "default" contract build-level and continuation-mode should be implementation-defined

Abstract

The current WP is trying to overreach in its specification of what a default build-level for contract attributes is, and what the default setting for continuation mode is. That overreach either has no meaning, or it's potentially dangerously wrong, and we shouldn't try to limit implementations with either meaningless specifications or specifications that try to dictate to implementations what kind of configuration interfaces they provide for programmers. We cannot know what sort of configuration interfaces and with what defaults are appropriate, so the solution is to remove suggestions on that note from the specification. Furthermore, attempts to turn C++ into a "safe" language by default are questionable.

Rationale

First of all, practically:

```
clang++ foo.cpp
```

does not compile `foo.cpp` as a C++ program. It compiles it with extensions enabled. Any lofty attempts to make compilers do what-ever by default because the standard says so are foolish.

Second, again practically:

```
g++ -O2 foo.cpp
```

or even

```
g++ -pedantic -O2 foo.cpp
```

has a well-established meaning, and enabling contract checks there would be a disturbing change.

An implementation could just skate around by our attempts to enable contract checking by default. There are (at least) two ways they can do it:

1. Impl-define that any mention of an optimization flag is a contract build level selection (but some of us wouldn't want that, either).
2. Impl-define that the actual C++ compiler (cc1plus) is the C++ compiler, not the front-end (g++), and orchestrate things so that an invocation of the front-end always passes a `-fdisable-contracts` down to the `c++` compiler, and then our words about default modes mean nothing.

All that is, to some extent, beside the more important point. We cannot know whether contract-checking-by-default is a good choice for C++ at large, in general. We do not have any deployment experience to give us the confidence to go to that direction. Therefore what we should do is leave such decisions to the tools, instead of trying to dictate a lofty doctrine we have no confidence in.

Wording

In [dcl.attr.contract.check]/3, modify as follows:

~~If no build level is explicitly selected, the build level is default.~~ The mechanism for selecting the build level is implementation-defir

In [dcl.attr.contract.check]/7, modify as follows:

A translation with a violation continuation mode set to on continues execution after completing the execution of the violation handler. ~~If no continuation mode is explicitly selected, the default continuation mode is off.~~